Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/30/2026
PI Name:	Allen, Jeffrey S.		
Project Title:	Reduced-Order Modeling of Interfacial Dynamics to Enable Large-Scale, Mission-Length Simulations of Low-Gravity Propellant Management Using CVB PSI Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	49931	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	10/01/2023	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per J. McQuillen/GRC (Ed., 9/29/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bellur, Kishan Ph.D. (University of Cincinnati)		
Grant/Contract No.:	80NSSC24K0193		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current state-of-the-art allows for simulation of liquid-vapor phase change and thermal management of large (1 to 10 meter diameter), low-gravity orbiting propellant depots (LH2 and LOx) for a few seconds with weeks-to-months of computational time. Part of the computational cost arises from having to resolve microscale features (10e-6 meters) on a dynamic liquid-vapor interface (ullage waves and liquid slosh) in order to capture local phase change and heat transfer. The result is an extremely fine mesh on a very large computational domain. The objective of the proposed work is to develop a reduced-order interface model capable of capturing all the relevant physics of interfacial dynamics for two-phase flow in low gravity. We intend to expand a non-linear evolution of the interface using a Eulerian vortex sheet model implemented on an unfitted finite element mesh. The interfacial model will be formulated to integrate mass, momentum, and energy exchange at the interface into an evolution equation for a vortex sheet. The unique configuration, test conditions, and data (imaging and thermal) of the Constrained Vapor Bubble (CVB) experiment conducted on the International Space Station (ISS) can isolate several important instability physical mechanisms, such as Marangoni stresses. CVB data, without the overpowering effect of gravity, will enable incorporation and validation of the nuances of unstable interfacial dynamics to the reduced-order interface model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The new computational method for modeling an evaporating and condensing interface will advance our understanding of cryogenic hydrogen storage, which will benefit the overall hydrogen economy.
Task Progress:	An Eulerian vortex sheet evolution equation was solved on an implicit level set representation of the interface. Since Task 1 did not involve any embedding technique, the focus was solely on formulating the vortex sheet evolution equation. The model comprised of a single global mesh for the entire computational domain. Accurately capturing the effects of the changing shape of a moving interface in most reduced order models requires representing the curvature and the derivative of curvature. Therefore, it became necessary to use a very fine mesh with a high order numerical scheme. The simulation is of a Bubble, Drop, and Particle Unit (BDPU) experiment in which a gas bubble moving across a temperature gradient due to Marangoni flows. A dense mesh along the entire path of the bubble, or a continuous re-meshing of the domain, is required for the level set function to accurately capture the interface curvature and its derivative as the interface moves. This resulted in large global matrices needing to be solved at every time step, making the solution time impractical. The initial Eulerian representation, coupled with a level set solution, did show promising results, but due to slow convergence it was impossible to realistically validate the solution against experiments. It was decided to approach this problem from a Lagrangian perspective and restrict the solution to the interface boundary. We are using an Arbitrary Lagrangian-Eulerian Formulation (ALE) to solve this problem, using the Biot-Savart law to calculate interface velocity. The ALE approach was orders of magnitude faster than the Eulerian Level-Set formulation. However, it was easier to implement the effects of viscous dissipation on vortex sheet strength using a level set formulation.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Rajan A, Allen JS, Bellur K. "Reduced-order modeling of low gravity liquid-vapor interfacial dynamics using a vortex sheet evolution equation." 2025 Midwestern Universities Fluid Mechanics (MUFMech) Retreat, Rochester, Indiana, Apr 10-12, 2025. Abstracts. 2025 Midwestern Universities Fluid Mechanics (MUFMech) Retreat, Rochester, Indiana, Apr 10-12, 2025. , Apr-2025
Articles in Peer-reviewed Journals	Chakrabarti U, Allen JS, Bellur K. "Investigating the inequality of phase change coefficients using ISS experimental data." Nanoscale and Microscale Thermophysical Engineering. 2025 Jul 3;29(2-3):139-58. https://doi.org/10.1080/15567265.2025.2544877 , Jul-2025
Papers from Meeting Proceedings	Rajan A, Allen JS, Bellur K. "Reduced-order modeling of low gravity liquid-vapor interfacial dynamics using a vortex sheet evolution equation." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Papers from Meeting Proceedings	Ragan A, Allen JS, Bellur K. "A comparative analysis of a reduced order vortex sheet model of the liquid-vapor interface in microgravity against existing interface tracking and interface capturing methods." 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. Abstracts. 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. , Dec-2025